

Kaizen Event - Meeting Notes (Raw)

Continuous Improvement Workshop / Internal working document / not yet summarised

Event title	Reduce changeover time and start-up waste on Line 4 (laminated dough / croissant line)
Type	3-day Kaizen event (SMED + waste reduction focus)
Dates	Tuesday to Thursday, week 18
Location	Bakery hall, Line 4 area + CI room
Facilitator	CI Engineer
Sponsor	Plant Manager
Scope line	Line 4 only. Mixing and final logistics out of scope for this event.

Attendees

Present across the three days (not everyone every session): Plant Manager (sponsor, opening and closing only), CI Engineer (facilitator), Line 4 Shift Leader (day), Line 4 Shift Leader (night, day 1 and 3), two operators from Line 4, Maintenance Technician (mechanical), Maintenance Technician (electrical, day 2), Quality Coordinator, Planning Coordinator, Process Engineer, and an HR Business Partner who joined the day 1 kick-off for the people side. Apologies from the Packaging Lead who was on holiday but sent notes.

Why we ran this event (background and problem)

Line 4 is our newest laminated dough line and was commissioned about ten months ago. It is strategically important because it carries the growth volume for the retail private-label croissant and pain-au-chocolat range. However, since ramp-up the line has consistently underperformed against its OEE target. The headline number is that Line 4 is running at roughly 58 percent OEE against a target of 75 percent, and when we broke that down the single largest loss is not breakdowns or speed, it is changeover and the associated start-up waste. We change product on Line 4 between four and seven times per day depending on the schedule, and each changeover is taking far longer and generating far more scrap dough than we assumed in the business case. Planning have started to avoid scheduling short runs on Line 4 because the changeovers are so painful, which means we are losing the flexibility the line was supposed to give us. The Plant Manager asked us to run a focused Kaizen rather than keep discussing it in production meetings.

Event charter and objectives (agreed day 1 morning)

- Reduce average changeover time on Line 4 from the current baseline to a target that makes short runs viable.
- Reduce start-up waste (scrap dough and out-of-spec product at run start) generated per changeover.
- Standardise the changeover so it does not depend on which shift leader is on.
- Do this without capital investment where possible. Quick wins first.

Current state - what we measured (day 1)

We timed five real changeovers on day 1 and pulled the historical data from the line terminal for the previous month. The numbers were worse and more variable than people expected. Average measured changeover time door-to-door (last good product of run A to first good product of run B) was 47 minutes. The fastest we saw was 31 minutes and the slowest was 68 minutes for what was nominally the same type of changeover. That variation alone tells us the process is not standardised. Start-up scrap averaged about 42 kg of dough per changeover, with the worst case at over 70 kg, mostly from the first products

through the lamination and forming section before the line stabilises. The night shift was on average noticeably slower than the day shift, and when we asked why, it became clear they have less support and tend to do tasks in series that the day shift does in parallel.

Rough breakdown of an average 47-minute changeover (from the timing study):

Step	Avg time	Observation
Stop, clear line, remove run-A product	8 min	Mostly necessary, some waiting
Clean forming/lamination contact parts	11 min	Done in series, one person, no kit prepared
Change forming tooling / cutting set	9 min	Tools fetched from store mid-changeover (avoidable)
Adjust proofer and oven settings	6 min	Manual, by memory, trial and error
Re-thread / reload packaging film if needed	5 min	Sometimes skipped, sometimes not
Start up, dial in, reach good product	8 min	Where most scrap is generated

Observations from going to the floor (gemba walk, day 1 afternoon)

We physically watched two changeovers and noted everything. Key observations that everyone in the room recognised once it was written down: there is no changeover checklist or standard work, so each person does it their own way and from memory. The forming tooling and the cutting set are stored in a cabinet about forty metres from the line, and nobody fetches them before stopping the line, so a chunk of the changeover is just walking. The proofer and oven parameters for each product are not saved as recipes that can be recalled with one button, even though the control system can do this, so the operators re-enter or adjust them manually each time, which is slow and is also a source of the start-up scrap because the first products go through before conditions are right. Cleaning materials and the few special tools needed are not staged, so people leave the line to get a scraper or a cloth. There was also a clear internal-versus-external setup issue: a lot of activity that could happen while the previous run is still going (preparing tooling, pre-staging the next recipe, getting materials ready) is currently done after the line is stopped, which is exactly the SMED problem.

Root cause discussion (day 2 morning, fishbone + 5 whys)

We built a fishbone around the question why is the changeover slow and variable, across method, machine, people, material. The recurring themes that came out: Method - no standard, internal and external setup not separated. Machine - recipes exist in the control system but are not configured or used; tooling change is more manual than it needs to be. People - knowledge sits in a few experienced heads, new and agency operators have not been trained on a standard because there is not one to train on. Material - tools and consumables not staged at point of use. A 5-whys on the start-up scrap specifically landed on: we generate scrap because the first products are out of spec, because the proofer and oven are not at the right condition at start, because settings are entered manually and by trial and error, because the saved-recipe function was never set up during commissioning, because at ramp-up everyone was focused on getting the line running at all and the changeover optimisation was never closed out. That last point was an honest moment for the room.

Countermeasures brainstormed and selected (day 2 afternoon)

We generated a long list and then selected based on impact versus effort. Selected actions:

- Create a standard changeover sheet (standard work) with photos, agreed during the event by day and night shift leaders together.
- Separate internal and external setup: define everything that can be prepared before the line stops and build it into the standard.

- Set up and save proofer and oven recipes per product in the control system so settings are recalled with one selection, not re-entered.
- Build a changeover trolley / shadow board staged at the line with the tooling, cutting sets, cleaning kit and consumables needed.
- Train all Line 4 operators including agency staff on the new standard, and capture sign-off (note: links to the audit training-record gap).
- Move tooling storage from the distant cabinet to a point-of-use location.

Actions tried live during the event (day 3 morning)

Rather than just plan, we ran two changeovers on day 3 using a first-draft standard, with tooling pre-staged on a borrowed trolley and with the maintenance technician having pre-configured three product recipes in the control system the evening before. The first attempt came in at 29 minutes and the second at 24 minutes, against the 47-minute baseline. Start-up scrap on the second run was around 16 kg versus the 42 kg average, mainly because the oven and proofer were already at the saved condition when the line started. The operators commented that having the trolley there and not walking to the store made the biggest felt difference, and the recipe recall removed the guesswork. We did not change any hardware. These are not validated long-run numbers but they show the direction is right.

Results summary (preliminary, to be confirmed over 4 weeks)

Metric	Before (baseline)	After (event trials)	Target
Avg changeover time	47 min	24 to 29 min	< 25 min
Changeover variation (range)	31 to 68 min	24 to 29 min	tight / standardised
Start-up scrap per changeover	~42 kg	~16 kg	< 15 kg
Standard work exists	No	Draft v1 created	Approved + trained
Recipes saved in control system	0	3 of ~12 products	All products

Action list / follow-up (owners and dates agreed day 3 close)

#	Action	Owner	Due
1	Finalise and laminate the standard changeover sheet with photos	Day Shift Leader + CI Engineer	End of week 18
2	Configure and save remaining product recipes in control system	Process Engineer + Maintenance	End of week 20
3	Procure and build the dedicated changeover trolley / shadow board	Maintenance (mech)	Week 21
4	Relocate forming tooling to point-of-use storage at Line 4	Maintenance (mech)	Week 20
5	Train all Line 4 operators + agency on new standard, capture sign-off	Day Shift Leader + HR BP	Week 21
6	Confirm sustained results over 4 weeks and update OEE tracking	CI Engineer	Week 23
7	Report results to plant leadership and decide on rollout to Line 2	Plant Manager (sponsor)	Week 23

Open points and risks noted in the room

- Trial numbers are not yet validated over a sustained period; night shift not yet trained on the standard.
- Recipe configuration depends on maintenance availability, which is already stretched (open work orders elsewhere).
- Some scepticism on the floor that the standard will be followed once the event energy fades; sustainment plan needed.

- If results hold, there is a strong case to replicate on Line 2, but that needs a separate slot.

Closing comment from sponsor

The Plant Manager closed by thanking the team, noting that this is exactly the kind of structured problem solving we want more of, and asked for a single clear one-page summary that can be shown to regional leadership and used to justify replicating the approach on other lines. That one-pager is what we now need to produce from these notes.